

ALEJANDRO ITO ARAMENDIA

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EDUCATION

Tohoku University, Japan – BEng Mechanical and Aerospace Engineering Oct 2021 – Sept 2025

- **GPA:** 3.0 (UK 2:1 Equivalent)
- **Relevant Coursework:** Finite Element Method Analysis (Python) (90%), Pygame (Python, Object-Oriented Programming), Data Structures and Algorithms (90%), Information Processing (C++) (90%)

King's College London Maths School, UK – A-Levels Sept 2019 – Aug 2021

- A*AAA in Maths, Further Maths, Physics and Computer Science (AS)

EXPERIENCE

Research Assistant (Software Developer) – Ishii Lab, Tohoku University Oct 2023 – Sep 2024

- Developed a MATLAB GUI to automate complex vector fluid flow analysis, increasing data processing efficiency for research teams and enhancing multi-dataset analysis by over 300%.
- Applied bilinear interpolation and regression models to analyse 10+ tissue flow sequences, enabling more precise speed/direction mapping for clinical ultrasound research.

International PBL Workshop 2023 – National Taipei University of Technology Aug 2023 – Aug 2023

- Engineered an Arduino robot in C++ to detect and transport coloured objects with 90%+ accuracy, integrating real-time sensor feedback and object recognition algorithms.
- Took the lead in team presentations, clearly communicating project goals and updates to a panel of judges and 100+ participants.

PROJECTS

Japanese Learning Platform (Full-Stack Development) (In Progress) Dec 2024 – Present

- Developing a responsive, full-stack web application leveraging Next.js (React) for frontend, and Node.js backend services with RESTful APIs, hosted on AWS Lambda.
- Architecting backend infrastructure with AWS RDS and AWS S3 for scalable file and data handling.

Monte Carlo Percolation Simulator (Java + Python) Mar 2025 – Mar 2025

- Built a multithreaded Java simulation to estimate the percolation threshold using Union-Find and Monte Carlo methods, with performance-optimised logic and concurrency via ExecutorService.
- Built an interactive Swing GUI with dark mode, grid controls and speed adjustment, Dockerized for cross-platform deployment, and integrated Python for data visualisation.

Finance Tracker (Backend Development) Nov 2024 – Dec 2024

- Implemented a fully serverless REST API using Node.js, AWS Lambda, and API Gateway, implementing secure JWT-based authentication and CRUD operations for finance data.
- Integrated a comprehensive CI/CD pipeline (GitHub Actions) with automated unit tests (Jest), ensuring high code quality and rapid, reliable deployments to AWS.

Technical Writer on Machine Learning and Algorithms (2000+ Likes) Jan 2024 – May 2024

- Published 10+ articles on deep learning and data structures, gaining 2000+ likes on Medium.

CERTIFICATIONS

Accenture UK Developer & Technology Virtual Experience – Forage Mar 2025 – Mar 2025

- Completed a project-based simulation of the SDLC, Agile/Waterfall and DevOps workflows, solving 5+ technical tasks across debugging, algorithm design and testing.

SKILLS

Languages/Frameworks: JavaScript, Java, Python, SQL, MATLAB, Node.js, Express, React.js, Next.js, HTML, CSS

Databases: PostgreSQL, MongoDB, Redis

Cloud/DevOps: AWS (Lambda, API Gateway, RDS, S3), Docker, Git, CI/CD Pipelines, Linux, Bash, Postman, Agile

Testing/Tools: Jest (Unit Testing), Bcrypt, JWT, RESTful APIs, Pandas

Other: Multithreading, Object-Oriented Programming, Data Structures & Algorithms, Data Visualisation

Languages: English (Native), Spanish (Advanced), Japanese (JLPT N2)